HY-World 2.0: A Multi-Modal World Model for Reconstructing, Generating, and Simulating 3D Worlds

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https://arxiv.org/abs/2604.14268

Research Questions

- How can world generation and 3D reconstruction be unified in a single multi-modal framework?
- How can consistent memory mechanisms enable high-fidelity 3D world expansion from sparse inputs?
- How can feed-forward 3D reconstruction models be improved for universal geometric prediction from multi-view inputs?

Core Idea

HY-World 2.0 addresses the bifurcation in 3D world modeling where existing methods either excel at generation from sparse inputs or reconstruction from dense multi-view data, but not both. The key insight is a four-stage pipeline that bridges video diffusion priors with geometric 3D consistency: panorama generation creates 360° scene initialization, trajectory planning determines optimal camera paths, world expansion uses memory-guided keyframe generation to maintain consistency, and world composition reconstructs final 3DGS assets. The memory mechanism in WorldStereo 2.0 is crucial - it maintains global geometric memory and improved spatial-stereo memory to ensure multi-view consistency during keyframe generation, while WorldMirror 2.0 provides robust feed-forward reconstruction that works for both generated and real multi-view inputs.

Key Formula

Normalized Position Encoding

$$\tilde{x}_i = \frac{2i+1}{H_p} - 1, \quad \tilde{y}_j = \frac{2j+1}{W_p} - 1$$

Maps patch positions (i,j) to normalized coordinates [-1,1] for resolution-independent position encoding. H_p and W_p are patch grid dimensions. This enables flexible multi-resolution inference by converting position extrapolation into interpolation.

Depth-to-Normal Loss

$$\tilde{N}_i(x) = \text{normalize}\left(\frac{\partial P_i}{\partial u} \times \frac{\partial P_i}{\partial v}\right), \quad P_i = K^{-1} D_i \cdot [u, v, 1]^T$$

Converts predicted depth D_i into surface normals via back-projection and cross products. K is camera intrinsics matrix, u,v are pixel coordinates. This provides explicit geometric supervision coupling depth and normal predictions for improved 3D consistency.

Key Figure

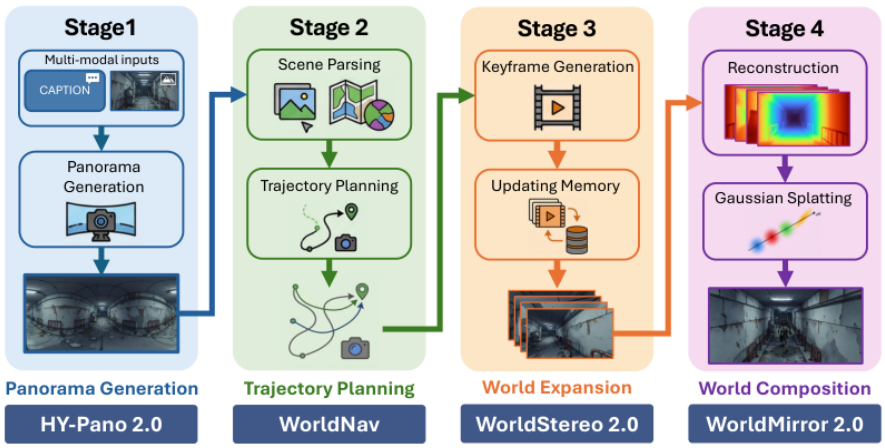

Figure 2: **Architecture of HY-World 2.0.** Our framework presents a four-stage process to transform multi-modal inputs into immersive 3D worlds: (1) initializing the world via *Panorama Generation*, (2) deriving exploration camera paths through *Trajectory Planning*, (3) expanding the world observations via memory-driven *World Expansion*, and (4) constructing the final 3DGS assets using *World Composition*. The core model/algorithm used in each stage is denoted at the bottom.

Figure 2: Shows the complete four-stage architecture that unifies world generation and reconstruction in a single framework.

Limitations

- Requires substantial computational resources for the four-stage pipeline with multiple large models
- Memory mechanisms may struggle with very long trajectories or complex scene dynamics
- Depth alignment process relies on geometric guidance that may fail in textureless or reflective regions
- Performance depends heavily on quality of initial panorama generation which may fail for complex prompts